

# Propagating Reliability Through Neural Computation

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## Abstract

Standard neural layers propagate representations without explicitly preserving how strongly those representations are supported by the available observations. We introduce the Belief-Valued Unit (BVU), a deterministic neural unit that propagates content, inherited support, and representation conflict. These quantities define a computational precision that influences both aggregation and activation, while satisfying structural properties including conservative precision propagation and replication invariance. Across seven clean supervised-learning benchmarks, BVU is broadly competitive with conventional networks but shows no consistent universal advantage. Under controlled missing-feature and heterogeneous-noise corruption, however, BVU achieves higher corruption AUC than a parameter-matched reliability-aware MLP in seven of eight stable dataset-corruption comparisons. On MNIST and Fashion-MNIST, the advantage persists against a same-width baseline with approximately  $1.65\times$  as many parameters. On APS Failure, BVU is competitive with NeuMiss overall (0.844 versus 0.838 PR-AUC) and separates most strongly among examples with more than 50% missing features (0.848 versus 0.702). On Air Quality, BVU reaches  $R^2 = 0.760$  versus 0.743 for NeuMiss. Hidden computational precision responds systematically to information degradation, and shuffling feature-reliability alignment on APS reduces PR-AUC by 0.057. A non-neural boosting reference remains stronger on APS. These results support reliability propagation as a specialized inductive bias for heterogeneous-quality information rather than a universal replacement for conventional neural computation.

## 1 Introduction

Neural networks typically propagate representations through sequences of scalar activations. Once an input has been mapped into a hidden representation, downstream layers receive the representation itself but not an explicit description of how strongly that representation is supported by the underlying observations. In many learning problems this distinction is unnecessary: the available inputs are treated as equally valid, and the network need only learn a useful representation. In other settings, however, the quality of the available information can vary substantially across features and examples. Sensor measurements may be missing, individual features may be corrupted by different levels of noise, and some observations may be substantially less reliable than others.

A common solution is to expose this information to the model explicitly, for example through missingness indicators, confidence values, or reliability features. Such information can help a conventional network learn how to treat unreliable observations. This raises a more basic architectural question: *should reliability be treated only as another input feature, or should it remain part of the computation as representations are transformed through the network?*

We study the latter possibility. We introduce the *Belief-Valued Unit* (BVU), a neural unit whose hidden state contains three quantities,

$$B = (\mu, e, u), \quad (1)$$

representing content  $\mu$ , inherited support  $e$ , and conflict  $u$  generated by disagreement among the representations contributing to the unit. Support and conflict determine a computational precision

$$\pi = \frac{e}{1 + eu}, \quad (2)$$

which controls both how strongly incoming representations contribute to the next hidden state and how strongly the resulting consensus is expressed by the unit.

The resulting computation differs from simply attaching a confidence value to the network input. Reliability is recomputed recursively. A unit first forms a precision-weighted consensus of its incoming representations, then measures disagreement around that consensus, and finally combines the inherited support and newly generated conflict into the precision of its output state. The state  $(\mu, e, u)$  is then passed to the next belief-valued layer. The additional quantities are internal computational variables rather than estimates of calibrated predictive uncertainty.

We design the unit so that this additional state obeys several simple structural properties. Output precision cannot exceed the most precise input, duplicating identical evidence does not artificially increase precision, and uniformly lowering source reliability reduces output precision even when the relative reliability of the sources is unchanged. Under neutral reliability and zero conflict, the computation reduces to an ordinary linear-tanh neural unit. The layer can also be expressed using standard dense matrix multiplications, preserving the asymptotic scaling of a dense feed-forward layer up to a constant-factor computational overhead.

Our experiments examine whether this additional machinery is useful in the regime for which it is intended. On standard clean supervised-learning benchmarks, the proposed unit is generally competitive with conventional networks but provides no consistent universal advantage. Under controlled missing-feature and heterogeneous-noise corruption, however, the behavior changes. The proposed model outperforms a parameter-matched MLP supplied with the same feature-level reliability information in seven of eight stable dataset-corruption comparisons. On MNIST and Fashion-MNIST, the advantage persists when the conventional reliability-aware MLP is given the same hidden width and approximately  $1.65\times$  as many trainable parameters.

We further evaluate the method on two datasets with naturally missing sensor measurements. On APS Failure at Scania Trucks, the proposed model is competitive with NeuMiss on the complete test set, while its relative advantage becomes substantially larger among examples with the highest fractions of missing features. On UCI Air Quality, it obtains the strongest mean  $R^2$  among the evaluated neural models. At the same time, a non-neural gradient-boosting reference remains stronger on APS, and several negative results show that the proposed unit should not be interpreted as a general-purpose replacement for conventional neural computation.

The internal state exhibits behavior consistent with the intended mechanism. Hidden precision decreases systematically as synthetic corruption increases. On APS, the fully observed examples form a distinct regime, while among examples containing missing inputs, hidden precision decreases as missingness becomes more severe. In addition, shuffling the alignment between features and their supplied reliabilities causes a substantial decrease in predictive performance, indicating that the model uses the correspondence between representation content and reliability rather than merely the marginal distribution of reliability values.

Taken together, the results support a narrower claim than universal superiority: explicitly carrying computational reliability through hidden representations can provide a useful inductive bias when the quality of the available information is heterogeneous.

Our main contributions are:

- We introduce a belief-valued neural unit that propagates content, inherited support, and representation conflict through ordinary feed-forward layers, with computational precision influencing both aggregation and activation.
- We establish structural properties of the resulting computation, including conservative precision propagation, replication invariance, sensitivity to absolute reliability, and reduction to a conventional neural unit in the neutral-reliability limit.
- We evaluate the unit under controlled heterogeneous reliability and naturally missing sensor data, using reliability-aware conventional networks, stronger capacity controls, and NeuMiss as neural

baselines. The results indicate that the benefit is concentrated in settings where information quality varies substantially rather than in clean supervised learning generally.

- We perform interventions on the propagated reliability state and analyze hidden precision across corruption and missingness levels, providing evidence that the internal reliability mechanism is actively used by the trained models.

## 2 Related Work

Our work lies at the intersection of learning with missing data, confidence propagation, and neural representations that carry information beyond a single scalar activation.

### 2.1 Learning with missing and unreliable inputs

A large literature studies supervised learning when some input features are missing or unreliable. A common strategy is to expose missingness information directly to the predictor, either through explicit indicators or through architectures specialized for incomplete observations.

NeuMiss derives a neural architecture for supervised learning with missing features from an approximation to the optimal predictor under several missing data mechanisms (Le Morvan et al., 2020). Its characteristic nonlinearity multiplies hidden representations by the missingness indicator, allowing the network to adapt its computation to the observed feature pattern. We use NeuMiss as a real-data baseline because it is specifically designed for prediction with missing inputs.

Related work in computer vision has also incorporated validity information directly into the forward pass. Partial convolution conditions each convolution on valid pixels only and propagates an updated validity mask through the network (Liu et al., 2018). These approaches demonstrate that information about which observations are available can be useful beyond the input layer.

Our setting is broader than binary missingness. We allow each input feature to carry a continuous reliability value and propagate an internal reliability state through ordinary feed-forward layers.

### 2.2 Confidence propagation

The closest conceptual line of work is normalized convolution and confidence propagation for sparse regression. Eldesokey et al. introduce normalized convolutional layers that combine sparse measurements according to associated confidence values and propagate confidence to subsequent layers (Eldesokey et al., 2019). Later work extends this framework to model input and output uncertainty for depth completion (Eldesokey et al., 2020).

These methods establish that confidence can be carried jointly with a representation rather than used only as an input feature. BVU shares this general motivation, but differs in the state that is propagated. In our unit, downstream computational precision depends not only on inherited support but also on conflict generated from disagreement among the incoming hidden representations. Reliability is therefore recomputed as representations change through depth.

A recent complementary approach, ChannelProp, propagates missingness and confidence channels through linear layers according to the magnitudes of the learned weights and combines them with confidence-aware activation functions (Sani et al., 2026). This provides another example of retaining reliability metadata throughout a network. Our method instead uses a fixed analytic hidden-state update in which support and representation conflict jointly determine the precision used for both aggregation and activation.

### 2.3 Probabilistic and uncertainty-aware hidden representations

Several neural architectures propagate richer states than deterministic scalar activations. Natural-Parameter Networks represent both weights and neurons using exponential-family distributions and propagate their

parameters through multiple layers (Wang et al., 2016). Such models provide uncertainty-aware hidden representations as part of a probabilistic treatment of the network.

BVU is not probabilistic in this sense. Its support, conflict, and precision variables do not parameterize a distribution over the hidden activation or the model weights. They are deterministic computational states used to control how representations are combined.

Evidential deep learning provides another important distinction. Sensoy et al. (2018) use evidence to parameterize a Dirichlet distribution over class probabilities, enabling a deterministic network to express predictive uncertainty. Our goal is different: the quantities propagated by BVU are used internally during hidden computation and are not trained or interpreted as calibrated predictive uncertainty.

## 2.4 Position of the present work

The preceding methods show that missingness, confidence, and uncertainty can all be incorporated into neural computation in useful ways. Our contribution is a compact deterministic unit that combines three operations within each hidden layer: reliability-weighted content aggregation, propagation of inherited support, and generation of new conflict from disagreement among incoming representations.

The resulting precision affects both the aggregation performed by the unit and the magnitude of its output activation. This makes the reliability state recursive: it is not only transported from the input, but can be reduced by conflict that emerges from intermediate hidden representations.

## 3 Belief-Valued Neural Units

Standard neural layers propagate a scalar activation from one layer to the next. We instead associate each hidden unit with a three-component state

$$B_j = (\mu_j, e_j, u_j), \quad (3)$$

where  $\mu_j \in \mathbb{R}$  denotes the unit’s content,  $e_j > 0$  denotes inherited support, and  $u_j \geq 0$  denotes conflict accumulated during the current computation.

The quantities  $e_j$  and  $u_j$  are internal computational states rather than estimates of calibrated predictive uncertainty. Their role is to determine how strongly the representation carried by a unit should influence subsequent computation.

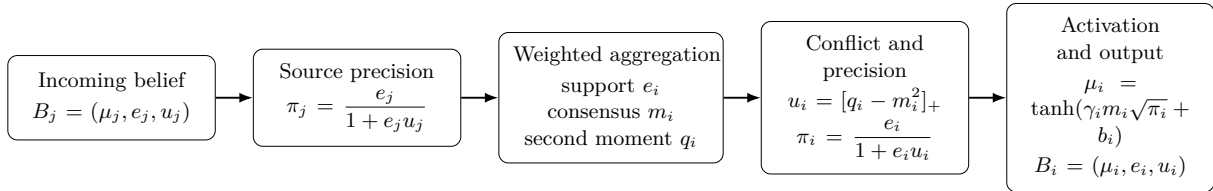


Figure 1: Overview of a belief-valued neural unit. Incoming content, support, and conflict determine source precision. The unit forms a reliability-weighted consensus, propagates support, generates new conflict from representation disagreement, and uses the resulting computational precision to modulate its output activation.

### 3.1 Computational precision

We combine support and conflict into an effective computational precision

$$\pi_j = \frac{e_j}{1 + e_j u_j}. \quad (4)$$

Equivalently,

$$\pi_j = \frac{1}{e_j^{-1} + u_j}. \quad (5)$$

Thus, lower inherited support or greater conflict reduces the influence of a unit.

At the network input, we initialize

$$\mu_j = x_j, \quad e_j = c_j, \quad u_j = 0, \quad (6)$$

where  $x_j$  is the input feature and  $c_j$  is an optional feature-level reliability value. When no reliability information is available, we set  $c_j = 1$ .

### 3.2 Normalized signed connectivity

Consider a layer with  $n$  input units and  $m$  output units. Let  $V \in \mathbb{R}^{m \times n}$  denote the trainable connection matrix. For output unit  $i$ , define the row magnitude

$$r_i = \sum_{k=1}^n |V_{ik}|, \quad (7)$$

We assume  $r_i > 0$ . In implementation, denominators are protected by a small numerical clamp to avoid division by zero; this does not affect the analysis below for nondegenerate rows.

We define unsigned and signed normalized coefficients

$$A_{ij} = \frac{|V_{ij}|}{r_i}, \quad S_{ij} = \frac{V_{ij}}{r_i}. \quad (8)$$

The unsigned coefficients  $A_{ij}$  are used when propagating support and measuring disagreement, while the signed coefficients  $S_{ij}$  determine how content is combined.

### 3.3 Support propagation and consensus

The support arriving at output unit  $i$  is

$$e_i = \sum_{j=1}^n A_{ij} \pi_j. \quad (9)$$

The corresponding precision-weighted signed consensus is

$$m_i = \frac{\sum_{j=1}^n S_{ij} \pi_j \mu_j}{e_i}. \quad (10)$$

Consequently, inputs with lower precision contribute less strongly to the content computed by the next unit.

### 3.4 Conflict from representation disagreement

In addition to the consensus, each unit computes a precision-weighted second moment

$$q_i = \frac{\sum_{j=1}^n A_{ij} \pi_j \mu_j^2}{e_i}. \quad (11)$$

We then define the output conflict as

$$u_i = [q_i - m_i^2]_+, \quad (12)$$

where  $[z]_+ = \max(z, 0)$ .

The quantity  $u_i$  is a disagreement statistic rather than a probabilistic variance. In particular, the consensus uses signed coefficients while the second moment uses their magnitudes. Its purpose is to detect disagreement among the representations contributing to a unit and reduce the precision of the resulting state.

The new output precision is therefore

$$\pi_i = \frac{e_i}{1 + e_i u_i}. \quad (13)$$

This construction separates two reasons for low computational reliability: insufficient inherited support, represented by  $e_i$ , and disagreement among the incoming representations, represented by  $u_i$ .

### 3.5 Reliability-modulated activation

The scalar content propagated to the next layer is

$$\mu_i = \tanh(\gamma_i m_i \sqrt{\pi_i} + b_i), \quad (14)$$

where  $b_i$  is a trainable bias and

$$\gamma_i = \text{softplus}(g_i) \quad (15)$$

is a positive trainable gain.

The resulting hidden state

$$B_i = (\mu_i, e_i, u_i) \quad (16)$$

is passed to the next belief-valued layer.

Precision therefore affects the computation twice: it determines how strongly incoming units participate in the consensus, and the resulting output precision modulates the activation generated from that consensus.

We initialize  $\gamma_i$  to approximately the  $\ell_1$  magnitude of row  $i$  of  $V$ . Under equal input precision and negligible conflict, this recovers the scale of an ordinary linear transformation before the nonlinearity.

### 3.6 Efficient implementation

The belief-valued layer can be evaluated using standard dense matrix operations without constructing explicit batch-output-input edge tensors.

For each output unit, define

$$L_i = \sum_j |V_{ij}|, \quad (17)$$

$$E_i = \sum_j |V_{ij}| \pi_j, \quad (18)$$

$$C_i = \sum_j V_{ij} \pi_j \mu_j, \quad (19)$$

$$Q_i = \sum_j |V_{ij}| \pi_j \mu_j^2. \quad (20)$$

The layer equations are equivalently

$$e_i = \frac{E_i}{L_i}, \quad (21)$$

$$m_i = \frac{C_i}{E_i}, \quad (22)$$

$$u_i = \left[ \frac{Q_i}{E_i} - m_i^2 \right]_+, \quad (23)$$

$$\pi_i = \frac{E_i}{L_i + E_i u_i}. \quad (24)$$

All four quantities can be computed using dense matrix multiplications; the two unsigned moments  $E$  and  $Q$  share the matrix  $|V|$  and may be evaluated together. An  $n$ -input,  $m$ -output layer contains  $mn + 2m$  trainable parameters and retains  $\mathcal{O}(Bmn)$  scaling for batch size  $B$ , with a constant-factor computational overhead over an ordinary dense layer.

Further implementation details and the vectorized formulation are provided in Appendix B.

## 4 Properties

The reliability state satisfies several structural properties independent of the training objective or dataset. We state the main results here and provide full proofs in Appendix A.

### 4.1 Conservative precision propagation

**Proposition 1.** For any nondegenerate output unit,

$$\pi_i \leq e_i \leq \max_j \pi_j. \quad (25)$$

Because  $e_i$  is a convex combination of incoming precisions and conflict is nonnegative, the computation cannot create greater precision than is available from its most precise input. Conflict can only reduce precision further.

### 4.2 Replication invariance

**Proposition 2.** If every input state and its associated connection weight are duplicated  $K$  times, the complete output state is unchanged:

$$(\mu'_i, e'_i, u'_i) = (\mu_i, e_i, u_i). \quad (26)$$

Thus, repeated copies of the same information cannot manufacture additional support or precision.

### 4.3 Sensitivity to absolute reliability

Suppose every incoming precision is multiplied by the same factor  $\lambda > 0$ . The consensus and conflict remain unchanged, while output precision becomes

$$\pi_i(\lambda) = \frac{\lambda e_i}{1 + \lambda e_i u_i}. \quad (27)$$

**Proposition 3.** For  $e_i > 0$ ,

$$\frac{\partial \pi_i(\lambda)}{\partial \lambda} = \frac{e_i}{(1 + \lambda e_i u_i)^2} > 0. \quad (28)$$

Uniformly lowering source reliability therefore reduces output precision even when the relative reliability of all sources is unchanged.

### 4.4 Positive row-scale invariance

**Proposition 4.** For fixed gain and bias, multiplying an entire connection row by any  $\alpha > 0$  leaves the output belief state unchanged.

The normalized coefficients  $A_{ij}$  and  $S_{ij}$  depend only on the relative pattern of the incoming weights, while the trainable gain  $\gamma_i$  separately controls activation scale.

#### 4.5 Ordinary-neuron limit

When all incoming states have neutral precision,

$$\pi_j = 1, \quad (29)$$

and the output conflict is zero, we have  $e_i = \pi_i = 1$ . If

$$\gamma_i = \sum_j |V_{ij}|, \quad (30)$$

then

$$\mu_i = \tanh \left( \sum_j V_{ij} \mu_j + b_i \right). \quad (31)$$

The belief-valued computation therefore contains a conventional linear-tanh unit as a special case rather than replacing ordinary neural computation with an unrelated operator.

### 5 Experimental Setup

We first evaluate seven standard classification and regression datasets: Breast Cancer Wisconsin (Diagnostic) (Wolberg et al., 1993), Wine (Aeberhard & Forina, 1992), Digits (Alpaydin & Kaynak, 1998), Diabetes (Efron et al., 2004), California Housing (Pace & Barry, 1997), MNIST (LeCun et al., 1998), and Fashion-MNIST (Xiao et al., 2017).

#### 5.1 Models and baselines

We compare BVU with four conventional neural controls.

**Plain MLP.** A parameter-matched multilayer perceptron receives only the observed or corrupted feature vector.

**Confidence MLP.** The model receives the concatenated input

$$[x, c], \quad (32)$$

where  $c$  is the same feature-level reliability supplied to BVU. Its total parameter count is matched to BVU as closely as possible.

**Same-width Confidence MLP.** To control for the reduction in hidden width caused by concatenating  $[x, c]$ , we additionally use a Confidence MLP with exactly the same hidden width as BVU, even when this gives the conventional model more parameters.

**Reliability-gated MLP.** For continuously varying reliability we also evaluate

$$x_{\text{gated}} = c \odot x, \quad (33)$$

testing whether applying reliability once at the input is sufficient.

For the real missing-data experiments, we additionally compare against NeuMiss (Le Morvan et al., 2020), using the authors’ implementation, and report HistGradientBoosting as a non-neural reference.

BVU initializes each input feature as

$$\mu_j = x_j, \quad e_j = c_j, \quad u_j = 0. \quad (34)$$

## 5.2 Clean supervised-learning benchmark

We first evaluate seven standard classification and regression datasets: Breast Cancer Wisconsin, Wine, Digits, Diabetes, California Housing (Pace & Barry, 1997), MNIST (LeCun et al., 1998), and Fashion-MNIST (Xiao et al., 2017).

Each dataset is evaluated using both 25% and 100% of the available training data and three random seeds. This benchmark tests whether the additional belief state changes ordinary supervised learning when heterogeneous reliability is not the central challenge.

## 5.3 Controlled heterogeneous reliability

We evaluate MNIST, Fashion-MNIST, Digits, and California Housing under two corruption families.

For missing-feature corruption, a feature is independently removed with probability  $p$  and assigned

$$c_j = \begin{cases} 1, & \text{observed,} \\ 10^{-3}, & \text{missing.} \end{cases} \quad (35)$$

Training uses

$$p \in \{0, 0.1, 0.2, 0.3\}, \quad (36)$$

while evaluation extends to

$$p \in \{0, 0.1, 0.3, 0.5, 0.7\}. \quad (37)$$

For heterogeneous Gaussian corruption,

$$\sigma_j \sim \text{Uniform}(0, s), \quad \tilde{x}_j = x_j + \epsilon_j, \quad \epsilon_j \sim \mathcal{N}(0, \sigma_j^2), \quad (38)$$

with reliability

$$c_j = \frac{1}{1 + \sigma_j^2}. \quad (39)$$

Training uses  $s \leq 0.75$ , while evaluation extends to  $s = 1.5$ .

Models are trained once per corruption family and evaluated across the full severity grid. We summarize robustness using the area under the performance-versus-corruption curve (corruption AUC).

## 5.4 Real missing-sensor data

We evaluate two datasets containing naturally missing measurements.

**APS Failure at Scania Trucks.** We preserve the official APS training and test sets (UCI Machine Learning Repository, 2016). Missing measurements are assigned reliability  $10^{-3}$  and observed measurements reliability 1. Because the task is highly imbalanced, PR-AUC is the primary metric. We also report the dataset’s asymmetric false-positive/false-negative cost in the appendix.

**Air Quality.** We predict carbon-monoxide concentration from sensor and environmental measurements in the UCI Air Quality dataset (Vito, 2008). Missing inputs are retained and the data are split chronologically into training, validation, and test partitions. The primary metric is  $R^2$ .

## 5.5 Training, interventions, and diagnostics

All primary results use three random seeds, identical data partitions, and identical synthetic corruption streams across competing models. Model selection uses validation data only.

For the real-data experiments, all neural models share the predeclared learning-rate grid

$$\{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}. \quad (40)$$

To test whether reliability is actually used, trained BVU models are evaluated without retraining under three conditions: true feature-aligned reliability, uniform reliability  $c_j = 1$ , and shuffled reliability that preserves the marginal reliability distribution while destroying its feature alignment.

We additionally record the internal precision

$$\pi = \frac{e}{1 + eu} \quad (41)$$

as corruption and naturally occurring missingness vary. For the real datasets, results are stratified by missing-feature fraction using the intervals 0, (0, 0.10], (0.10, 0.25], (0.25, 0.50], and > 0.50.

## 6 Results

We organize the results around three questions. First, does the proposed unit remain competitive when reliability is not a central feature of the task? Second, does recursively propagating computational reliability improve robustness when input quality varies? Third, does the same behavior appear on datasets containing naturally missing sensor measurements?

### 6.1 Clean-data behavior

On standard supervised-learning benchmarks without artificial corruption, BVU remains broadly competitive with its parameter-matched MLP comparison, but does not show a consistent universal advantage.

BVU has higher mean performance on MNIST and Fashion-MNIST at both training-data fractions and on California Housing at both fractions. Diabetes also shows positive mean differences, although the 25% condition has substantial seed variability. Breast Cancer, Wine, and Digits remain close to the matched MLP, with absolute mean differences below 0.01 in each training-data condition.

We therefore treat the clean benchmark primarily as a compatibility check: the additional belief state does not provide a general advantage when heterogeneous information quality is not central to the task. Complete per-dataset results are reported in Appendix D.1.

### 6.2 Robustness under heterogeneous input reliability

The behavior changes under controlled corruption. When different features are assigned different known reliabilities, BVU generally improves over a conventional MLP that receives the same reliability metadata through input concatenation.

Figure 2 shows performance as corruption severity increases on MNIST and Fashion-MNIST. The dashed vertical lines mark the largest corruption severity observed during training; points to the right therefore evaluate robustness outside the training corruption range.

Across the stable dataset-corruption combinations, BVU obtains higher corruption AUC than the parameter-matched Confidence MLP in 7 of 8 cases. The strongest and most consistent gains occur on MNIST and Fashion-MNIST, where BVU outperforms the Confidence MLP under both missing-feature and heterogeneous-Gaussian corruption for every random seed.

The improvement is not uniform across all datasets. In particular, Digits under missing-feature corruption remains a negative result, while the Gaussian-corruption comparison on Digits is approximately tied. California Housing also exhibits unstable conventional-baseline runs and is therefore not used as a headline robustness result.

Taken together, these results suggest that the advantage is associated with heterogeneous reliability rather than with clean supervised learning in general.

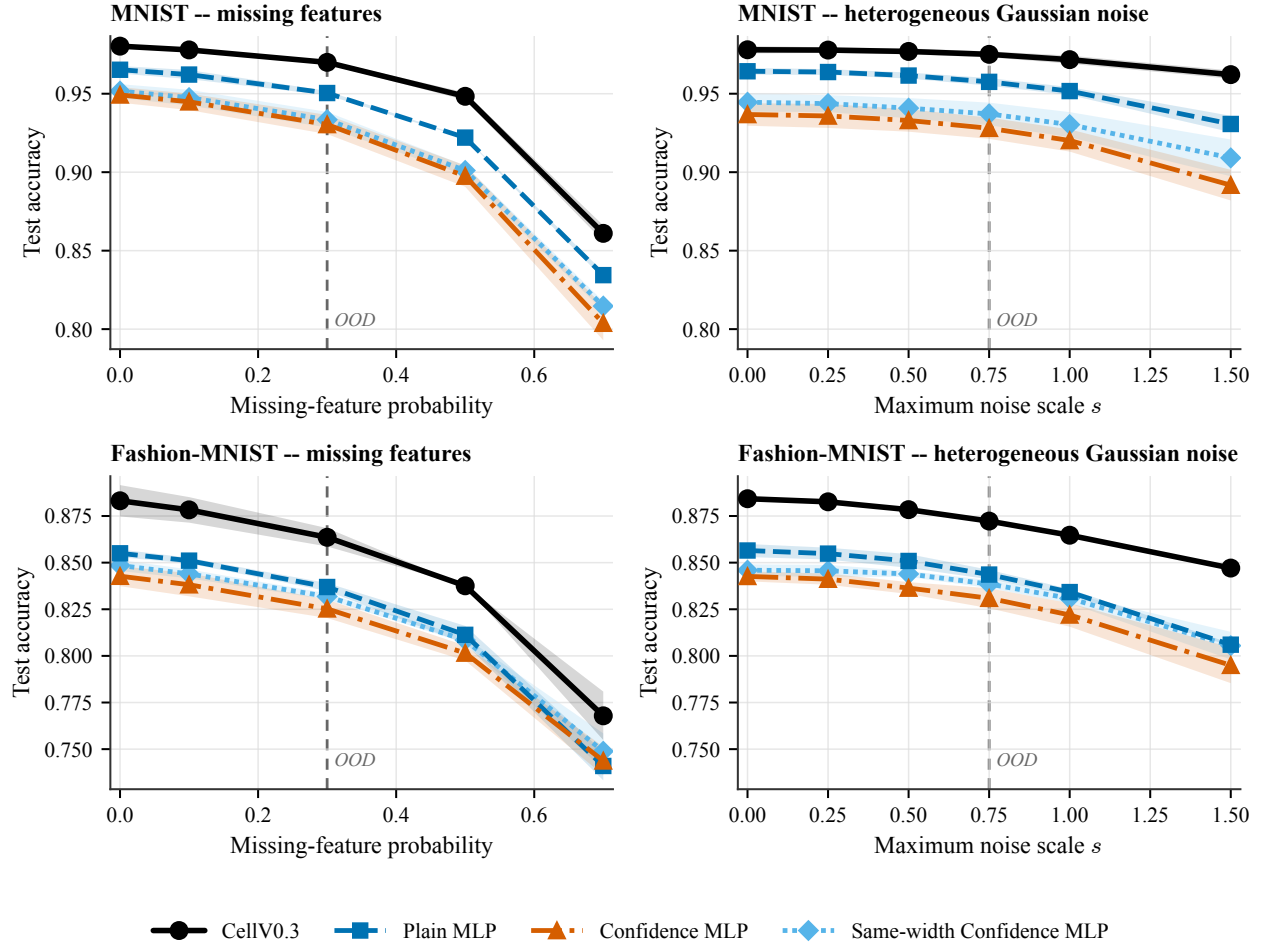


Figure 2: Robustness under controlled heterogeneous input reliability. Test accuracy is shown as missingness or noise severity increases on MNIST and Fashion-MNIST. BVU is compared with a Plain MLP, a parameter-matched Confidence MLP receiving the same reliability metadata, and a Same-width Confidence MLP that is allowed additional parameters. Dashed vertical lines indicate the largest corruption severity observed during training. Curves show mean performance over three seeds; shaded regions denote  $\pm 1$  standard deviation.

### 6.3 The robustness advantage is not explained by baseline capacity

The Confidence MLP receives the concatenated input  $[x, c]$ , which increases its input dimensionality and, under exact parameter matching, can reduce its hidden width. We therefore evaluated a stronger control using exactly the same hidden width as BVU, even when this gave the conventional model more parameters.

Table 1 reports corruption AUC for this experiment.

Table 2: Performance on real datasets containing naturally missing sensor measurements. Values are mean  $\pm$  standard deviation over three seeds for neural models. HistGradientBoosting is included as a non-neural reference.

| Model                     | APS PR-AUC                          | Air Quality $R^2$                   |
|---------------------------|-------------------------------------|-------------------------------------|
| Plain MLP                 | $0.826 \pm 0.016$                   | $0.750 \pm 0.016$                   |
| Confidence MLP            | $0.841 \pm 0.005$                   | $0.728 \pm 0.016$                   |
| Same-width Confidence MLP | $0.834 \pm 0.013$                   | $0.738 \pm 0.014$                   |
| BVU                       | <b><math>0.844 \pm 0.014</math></b> | <b><math>0.760 \pm 0.010</math></b> |
| NeuMiss                   | $0.838 \pm 0.012$                   | $0.743 \pm 0.011$                   |
| HistGradientBoosting      | <b><math>0.876 \pm 0.000</math></b> | $0.752 \pm 0.004$                   |

Table 1: Capacity-control experiment. The Same-width Confidence MLP uses the same hidden width as BVU and is allowed to exceed its parameter count.  $\Delta$  is BVU minus Same-width Confidence MLP corruption AUC.

| Dataset / corruption     | BVU   | Same-width | $\Delta$ | Param. ratio |
|--------------------------|-------|------------|----------|--------------|
| MNIST / missing          | 0.666 | 0.638      | +0.027   | $1.65\times$ |
| MNIST / Gaussian         | 1.460 | 1.400      | +0.060   | $1.65\times$ |
| Fashion-MNIST / missing  | 0.593 | 0.572      | +0.021   | $1.65\times$ |
| Fashion-MNIST / Gaussian | 1.305 | 1.251      | +0.054   | $1.65\times$ |
| Digits / missing         | 0.625 | 0.633      | -0.008   | $0.69\times$ |
| Digits / Gaussian        | 1.422 | 1.416      | +0.006   | $0.69\times$ |

On MNIST and Fashion-MNIST, the Same-width Confidence MLP contains approximately  $1.65\times$  as many parameters as BVU, yet BVU retains the higher corruption AUC in all four comparisons. The direction of the result is also consistent across all three random seeds in these four conditions.

The capacity control therefore rules out the most direct explanation that the earlier Confidence MLP results were caused by reducing the conventional model’s hidden width.

The Digits missing-feature result remains negative. In this case, the same-width model is actually smaller than BVU because a larger fraction of the latter’s parameter budget lies in its hidden-to-hidden transformation. We retain this result rather than treating it as an exception to the benchmark.

#### 6.4 Real missing-sensor data

We next evaluate whether the controlled-corruption behavior transfers to naturally occurring missing measurements. Table 2 reports the primary metric on APS Failure and Air Quality.

On APS, BVU and NeuMiss are close on the full test set:  $0.844$  versus  $0.838$  PR-AUC, respectively. The difference is small relative to the variation across seeds and should therefore be interpreted as competitive performance rather than a decisive overall advantage.

On Air Quality, BVU reaches  $R^2 = 0.760 \pm 0.010$ , compared with  $0.743 \pm 0.011$  for NeuMiss and  $0.738 \pm 0.014$  for the Same-width Confidence MLP.

The non-neural HistGradientBoosting reference achieves the best APS result, with PR-AUC  $0.876$ . We therefore do not claim state-of-the-art performance on tabular missing-data prediction. The comparison instead tests whether the proposed unit is competitive with strong neural approaches under naturally missing observations.

The aggregate APS scores, however, conceal substantial variation with the amount of missing information. We examine this dependence directly in Section 6.5.

### 6.5 Performance and internal precision vary with missingness

The full-test-set APS result hides a more structured pattern. Figure 3 stratifies test examples by the fraction of missing input features and compares predictive performance with BVU’s internal computational precision.

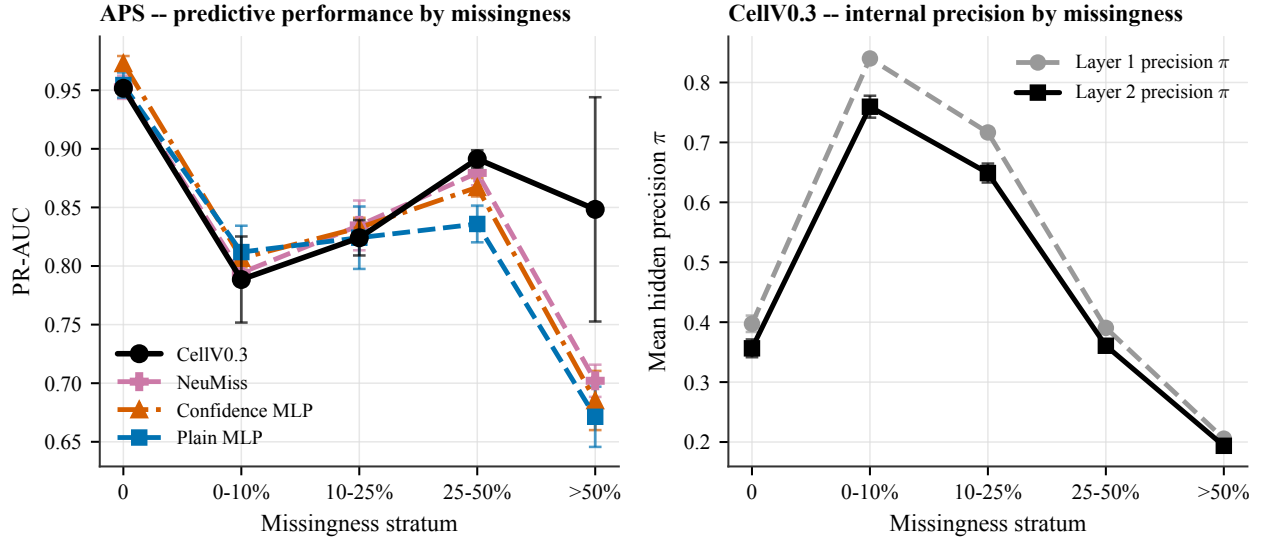


Figure 3: Performance and internal computational precision across naturally occurring missingness strata on APS Failure. Left: PR-AUC stratified by the fraction of missing input features. Right: BVU hidden computational precision over the same strata. The fully observed group forms a distinct precision regime; among examples with missing inputs, precision decreases as missingness becomes more severe. Curves show means over three seeds with  $\pm 1$  standard deviation.

As missingness increases, the relative advantage of BVU becomes more pronounced. For examples with more than 50% of features missing, BVU obtains PR-AUC 0.848, compared with 0.702 for NeuMiss and 0.671 for the Plain MLP. In the (0.25, 0.50] missingness interval, BVU obtains PR-AUC 0.891, compared with 0.879 for NeuMiss. At lower missingness levels, the neural models are substantially closer.

The internal-state pattern is more nuanced. Fully observed APS examples form a distinct precision regime. Among examples containing missing inputs, however, BVU’s hidden precision decreases consistently as the fraction of missing information increases and reaches its lowest value in the  $> 50\%$  missingness stratum. This behavior is not directly supervised: the training objective contains only the task-level classification loss.

The same qualitative relationship appears in the controlled-corruption experiments, where mean hidden precision decreases as corruption severity increases. On Air Quality, the mean second-layer precision decreases from approximately 0.41 for fully observed examples to approximately 0.001 in the highest-missingness group.

The joint pattern in Figure 3 is consistent with the intended mechanism: within incomplete examples, progressively less available evidence is accompanied by lower internal precision, while the relative benefit of explicitly carrying reliability is largest in the sparsest regime.

### 6.6 Reliability alignment matters

We next intervene on the reliability input at inference time while leaving the trained network weights and observed features unchanged.

On APS, shuffling the association between feature values and their reliability metadata reduces PR-AUC by approximately 0.057. Thus, preserving the marginal distribution of reliability values is insufficient; their alignment with the corresponding features matters to the trained computation.

However, replacing all APS reliabilities by one does not reduce performance. Relative to true reliability, the all-ones intervention changes PR-AUC by approximately  $-0.006$ , corresponding to a small improvement.

This result limits the strength of the mechanism claim. On APS, explicit reliability metadata is not necessary for good performance once the model has been trained, even though providing incorrectly aligned reliability is harmful.

On Air Quality, true reliability outperforms the all-ones intervention by approximately 0.012 in  $R^2$ . The shuffled-reliability intervention is not informative for this dataset because the missingness pattern makes the reliability vector effectively uniform within the relevant examples.

We therefore interpret the intervention results as evidence that the model can use feature-aligned reliability information, but not as evidence that explicit reliability metadata is universally required.

## 7 Discussion and Limitations

The experiments suggest that belief-valued computation is a specialized rather than universal inductive bias. BVU provides little consistent advantage when inputs are uniformly reliable, but becomes more useful when information quality varies substantially across features or when large fractions of the available evidence are absent.

### 7.1 When does reliability propagation help?

The strongest controlled results occur when different input features should not contribute equally to the computation. Under both missing-feature and heterogeneous-Gaussian corruption, BVU generally outperforms conventional networks given the same feature-level reliability metadata. On MNIST and Fashion-MNIST, the advantage survives a Same-width Confidence MLP with approximately  $1.65\times$  as many parameters.

The real-data experiments show a similar but more modest pattern. BVU and NeuMiss are close on the full APS test set, but their behavior separates more strongly among examples with high missingness. On Air Quality, BVU obtains the strongest mean  $R^2$  among the evaluated neural models.

These results suggest that the additional state is most useful when representation quality itself varies meaningfully. They do not support replacing conventional neural units indiscriminately in ordinary information-rich settings.

### 7.2 Interpreting the reliability state and its cost

BVU distinguishes representation content from the reliability of the computation producing that content. Support  $e$  summarizes inherited precision, while conflict  $u$  records disagreement generated during the current aggregation. Their combination

$$\pi = \frac{e}{1 + eu} \quad (42)$$

controls subsequent aggregation and activation.

These quantities should not be interpreted as calibrated predictive uncertainty. In particular,  $u$  is a deterministic disagreement statistic rather than predictive variance, and  $\pi$  is an internal computational precision rather than a probability of correctness.

The internal state nevertheless responds systematically to information quality. Under controlled corruption, hidden precision falls as corruption severity increases. On APS, fully observed examples form a distinct regime, while among examples containing missing inputs precision decreases as missingness becomes more severe. Shuffling the alignment between APS features and their reliabilities also degrades predictive performance.

This additional computation has a cost. The belief-valued layer propagates support and a second moment in addition to content, producing a constant-factor overhead over an ordinary dense layer. Although the vectorized implementation retains dense-layer asymptotic scaling, the additional cost is justified only when heterogeneous reliability is relevant to the task.

### 7.3 Limitations

Several results limit the scope of our conclusions.

BVU does not dominate every corruption benchmark; Digits under missing-feature corruption is a consistent negative example. On APS, the overall difference from NeuMiss is small relative to seed variation, and HistGradientBoosting outperforms all evaluated neural models. We therefore do not claim state-of-the-art tabular missing-data prediction.

The inference-time interventions also reveal an important limitation. Setting all APS input reliabilities to one does not reduce performance, even though shuffling their feature alignment is harmful. Explicit reliability metadata can therefore become redundant after training in some settings.

The reliability supplied at the input is known rather than learned. In the controlled benchmark it is specified by the corruption process, while for real missingness it is derived from whether a measurement is present. Future work could investigate learning reliability directly from raw observations.

The present real-data evaluation contains only two naturally incomplete sensor datasets, and the internal state remains deliberately low-dimensional. Support and conflict are each represented by one scalar per hidden unit and cannot capture arbitrary dependencies among uncertainty sources.

Finally, conflict can take large numerical values in severely sparse settings. In one APS seed, the mean conflict in the sparsest regime reached approximately  $1.6 \times 10^3$ . The resulting precision remained finite and no numerical divergence occurred, but the dynamic range warrants attention in larger or lower-precision implementations.

Overall, the results point toward a distinction between learning a representation and carrying information about how strongly that representation should participate in subsequent computation. Maintaining this distinction appears useful when information quality varies substantially, while conventional computation remains the simpler choice when it does not.

## 8 Conclusion

We introduced a belief-valued neural unit that propagates content together with inherited support and internally generated conflict. These quantities define a computational precision that influences both representation aggregation and the activation produced by each hidden unit. The resulting computation retains dense feed-forward scaling while satisfying simple structural properties, including conservative precision propagation and replication invariance.

Across clean supervised-learning tasks, the proposed unit provides no consistent universal advantage over conventional networks. Its benefit emerges instead when the quality of the available information varies. Under controlled missingness and heterogeneous corruption, BVU generally improves robustness over conventional networks supplied with the same reliability information, including stronger capacity controls. On naturally missing sensor data, it remains competitive with a specialized missing-data architecture, and its largest APS advantage occurs among examples with the highest missingness.

These findings suggest that explicitly carrying computational reliability through hidden representations can provide a useful inductive bias for learning from incomplete or heterogeneous-quality information. More broadly, they motivate treating representation content and representation reliability as distinct quantities when the quality of the available evidence is itself part of the learning problem.

## References

- Stefan Aeberhard and M. Forina. Wine. UCI Machine Learning Repository, 1992.
- E. Alpaydin and C. Kaynak. Optical recognition of handwritten digits. UCI Machine Learning Repository, 1998.
- Bradley Efron, Trevor Hastie, Iain Johnstone, and Robert Tibshirani. Least angle regression. *The Annals of Statistics*, 32(2):407–499, 2004. doi: 10.1214/009053604000000067.
- Abdelrahman Eldesokey, Michael Felsberg, and Fahad Shahbaz Khan. Confidence propagation through CNNs for guided sparse depth regression. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2019. doi: 10.1109/TPAMI.2019.2929170.
- Abdelrahman Eldesokey, Michael Felsberg, Karl Holmquist, and Michael Persson. Uncertainty-aware CNNs for depth completion: Uncertainty from beginning to end. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020.
- Marine Le Morvan, Julie Josse, Thomas Moreau, Erwan Scornet, and Gaël Varoquaux. Neumiss networks: Differentiable programming for supervised learning with missing values. In *Advances in Neural Information Processing Systems*, volume 33, 2020.
- Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998. doi: 10.1109/5.726791.
- Guilin Liu, Fitsum A. Reda, Kevin J. Shih, Ting-Chun Wang, Andrew Tao, and Bryan Catanzaro. Image inpainting for irregular holes using partial convolutions. In *Proceedings of the European Conference on Computer Vision*, 2018.
- R. Kelley Pace and Ronald Barry. Sparse spatial autoregressions. *Statistics & Probability Letters*, 33(3): 291–297, 1997.
- Naeem Shahabi Sani, Ferial Najiantabriz, Shayan Shafaei, and Dean F. Hougen. Evolving multi-channel confidence-aware activation functions for missing data with channel propagation. In *Proceedings of the Genetic and Evolutionary Computation Conference*, pp. 1011–1019. ACM, 2026. doi: 10.1145/3795095.3805092.
- Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- UCI Machine Learning Repository. APS Failure at Scania Trucks. UCI Machine Learning Repository, 2016.
- Saverio Vito. Air quality. UCI Machine Learning Repository, 2008.
- Hao Wang, Xingjian Shi, and Dit-Yan Yeung. Natural-parameter networks: A class of probabilistic neural networks. In *Advances in Neural Information Processing Systems*, volume 29, 2016.
- William Wolberg, Olvi Mangasarian, Nick Street, and W. Street. Breast cancer wisconsin (diagnostic). UCI Machine Learning Repository, 1993.
- Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-mnist: A novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.

## A Proofs of structural properties

### A.1 Proof of Proposition 1

By definition,

$$A_{ij} = \frac{|V_{ij}|}{\sum_k |V_{ik}|}. \quad (43)$$

Therefore

$$A_{ij} \geq 0, \quad \sum_j A_{ij} = 1. \quad (44)$$

Hence

$$e_i = \sum_j A_{ij} \pi_j \quad (45)$$

is a convex combination of the incoming precisions, which gives

$$e_i \leq \max_j \pi_j. \quad (46)$$

Since  $u_i \geq 0$ ,

$$\pi_i = \frac{e_i}{1 + e_i u_i} \leq e_i. \quad (47)$$

Combining the inequalities proves

$$\pi_i \leq e_i \leq \max_j \pi_j. \quad (48)$$

□

### A.2 Proof of Proposition 2

For one output unit define

$$L = \sum_j |V_j|, \quad (49)$$

$$E = \sum_j |V_j| \pi_j, \quad (50)$$

$$C = \sum_j V_j \pi_j \mu_j, \quad (51)$$

$$Q = \sum_j |V_j| \pi_j \mu_j^2. \quad (52)$$

Duplicating every source and corresponding weight  $K$  times gives

$$L' = KL, \quad E' = KE, \quad C' = KC, \quad Q' = KQ. \quad (53)$$

Therefore

$$e'_i = \frac{KE}{KL} = e_i, \quad (54)$$

and

$$m'_i = \frac{KC}{KE} = m_i. \quad (55)$$

Likewise,

$$u'_i = \left[ \frac{KQ}{KE} - (m'_i)^2 \right]_+ = u_i. \quad (56)$$

Hence  $\pi'_i = \pi_i$ , and the activation equation gives  $\mu'_i = \mu_i$ . □

### A.3 Proof of Proposition 3

Scaling every incoming precision by  $\lambda > 0$  gives

$$e'_i = \lambda e_i. \quad (57)$$

Because both numerator and denominator of the consensus scale by  $\lambda$ ,  $m_i$  remains unchanged. The same cancellation occurs in the second moment, so  $u_i$  is unchanged.

Thus

$$\pi_i(\lambda) = \frac{\lambda e_i}{1 + \lambda e_i u_i}. \quad (58)$$

Differentiating,

$$\frac{\partial \pi_i(\lambda)}{\partial \lambda} = \frac{e_i}{(1 + \lambda e_i u_i)^2} > 0 \quad (59)$$

whenever  $e_i > 0$ .  $\square$

### A.4 Proof of Proposition 4

Let

$$V'_{ij} = \alpha V_{ij}, \quad \alpha > 0. \quad (60)$$

Then

$$A'_{ij} = \frac{|\alpha V_{ij}|}{\sum_k |\alpha V_{ik}|} = A_{ij}, \quad (61)$$

and similarly

$$S'_{ij} = S_{ij}. \quad (62)$$

Consequently  $e_i$ ,  $m_i$ ,  $u_i$ , and  $\pi_i$  remain unchanged. For fixed  $\gamma_i$  and  $b_i$ , the activation  $\mu_i$  is therefore also unchanged.  $\square$

## B Implementation details

For a minibatch, let

$$P = \pi, \quad X = \pi \odot \mu, \quad Z = \pi \odot \mu^2. \quad (63)$$

The unnormalized moments can be evaluated as

$$E = P|V|^\top, \quad (64)$$

$$C = XV^\top, \quad (65)$$

$$Q = Z|V|^\top. \quad (66)$$

Because  $E$  and  $Q$  use the same matrix  $|V|$ , they may be evaluated jointly by stacking  $P$  and  $Z$  along the batch dimension before multiplication.

The implementation does not materialize tensors of shape  $\text{batch} \times \text{output} \times \text{input}$ . Numerical denominator clamps are used only to prevent division by zero in degenerate floating-point cases; the mathematical definitions in the main text assume nonzero support and nonzero weight-row magnitude.

## C Experimental details

### C.1 Controlled corruption

The missing-feature training severities are

$$p \in \{0, 0.1, 0.2, 0.3\}, \quad (67)$$

and evaluation uses

$$p \in \{0, 0.1, 0.3, 0.5, 0.7\}. \quad (68)$$

Missing standardized features are replaced with zero. Observed features receive reliability 1 and missing features receive reliability  $10^{-3}$ .

For heterogeneous Gaussian corruption,

$$\sigma_j \sim \text{Uniform}(0, s), \quad (69)$$

with

$$c_j = \frac{1}{1 + \sigma_j^2}. \quad (70)$$

Training uses

$$s \in \{0, 0.25, 0.50, 0.75\}, \quad (71)$$

and evaluation uses

$$s \in \{0, 0.25, 0.50, 0.75, 1.0, 1.5\}. \quad (72)$$

Synthetic corruption realizations are shared across competing models. Multiple corruption replicas are first averaged within model, seed, and severity before statistics are computed across seeds.

### C.2 APS Failure at Scania Trucks

The official training and test partitions are preserved. The official training set is divided into training and validation subsets using a deterministic stratified 80/20 split for each seed.

Preprocessing statistics are fitted on observed training values only. Observed features are standardized using the corresponding training mean and standard deviation, while missing standardized inputs are replaced with zero for MLP and BVU models.

Reliability is

$$c_j = \begin{cases} 1, & \text{observed,} \\ 10^{-3}, & \text{missing.} \end{cases} \quad (73)$$

The primary metric is PR-AUC. Secondary metrics include ROC-AUC, balanced accuracy, F1, precision, and recall. We also evaluate the official asymmetric cost

$$\text{Cost} = 10 \text{ FP} + 500 \text{ FN}. \quad (74)$$

The classification threshold used for this cost is selected on validation data and then frozen for test evaluation.

### C.3 Air Quality

Values encoded as  $-200$  are treated as missing. Rows whose target CO(GT) is missing are removed, while rows containing missing input measurements are retained.

The model uses the following eight inputs:

- PT08.S1(CO),
- PT08.S2(NMHC),
- PT08.S3(NO<sub>x</sub>),
- PT08.S4(NO<sub>2</sub>),
- PT08.S5(O<sub>3</sub>),
- temperature,
- relative humidity, and
- absolute humidity.

After target filtering, the chronologically ordered observations are split into the first 60% for training, the next 20% for validation, and the final 20% for testing.

Observed-feature standardization uses training statistics only. Missing standardized values are replaced with zero and assigned reliability  $10^{-3}$ ; observed values receive reliability 1.

The regression target is standardized using training-set statistics during optimization and converted back to its original scale for RMSE and MAE. The primary model-selection metric is validation  $R^2$ .

#### C.4 NeuMiss

We use the authors’ public NeuMiss implementation from `marineLM/NeuMiss_sota`, pinned to commit 7902b8d. Candidate depths are

$$\{1, 3, 5\}, \quad (75)$$

with depth selected using validation performance only.

#### C.5 Optimization and model configurations

All real-data neural models use AdamW and the shared learning-rate grid

$$\{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}. \quad (76)$$

Exact model widths, parameter counts, batch sizes, optimization limits, and checkpoint-selection settings are summarized in Tables 3, 4, 5, and 6.

Table 3: Clean benchmark model sizes for BVU and its parameter-matched MLP comparison.

| Dataset              | Model       | Hidden width | Params  | Batch  |
|----------------------|-------------|--------------|---------|--------|
| Breast Cancer        | BVU         | 83           | 9,879   | 85/128 |
| Breast Cancer        | MLP matched | 299          | 9,869   | 85/128 |
| Wine                 | BVU         | 90           | 9,903   | 26/106 |
| Wine                 | MLP matched | 582          | 9,897   | 26/106 |
| Digits               | BVU         | 123          | 24,733  | 128    |
| Digits               | MLP matched | 332          | 24,910  | 128    |
| Diabetes             | BVU         | 92           | 9,845   | 66/128 |
| Diabetes             | MLP matched | 829          | 9,949   | 66/128 |
| California Housing   | BVU         | 93           | 9,859   | 128    |
| California Housing   | MLP matched | 997          | 9,971   | 128    |
| MNIST; Fashion-MNIST | BVU         | 157          | 149,945 | 128    |
| MNIST; Fashion-MNIST | MLP matched | 187          | 148,675 | 128    |

Batch entries with two values are effective batch sizes for the 25% and 100% training fractions, respectively.

Table 4: Clean benchmark optimization and checkpoint settings. Identical model-family settings are collapsed.

| Dataset/group         |       |  | Optimizer | LR        | WD | Loss          | Max steps | Patience      | Selection metric |
|-----------------------|-------|--|-----------|-----------|----|---------------|-----------|---------------|------------------|
| Breast Cancer; Digits | Wine; |  | AdamW     | $10^{-2}$ | 0  | Cross-entropy | 15,000    | 1,500 (30×50) | Accuracy         |
| Diabetes              |       |  | AdamW     | $10^{-2}$ | 0  | MSE           | 15,000    | 1,500 (30×50) | $R^2$            |
| California Housing    |       |  | AdamW     | $10^{-2}$ | 0  | MSE           | 15,000    | 1,600 (8×200) | $R^2$            |
| MNIST; Fashion-MNIST  |       |  | AdamW     | $10^{-2}$ | 0  | Cross-entropy | 15,000    | 1,600 (8×200) | Accuracy         |

All clean rows use training fractions 25% and 100% and seeds 0, 1, and 2. Patience is shown in optimizer steps as validation checks × validation cadence. Maximum epochs were not recorded.

Table 5: Controlled-corruption and capacity-control model sizes.

| Dataset              | Model                            | Hidden width | Params  | Batch |
|----------------------|----------------------------------|--------------|---------|-------|
| MNIST; Fashion-MNIST | Plain MLP                        | 189          | 150,265 | 128   |
| MNIST; Fashion-MNIST | Confidence MLP                   | 95           | 150,015 | 128   |
| MNIST; Fashion-MNIST | Reliability-gated MLP            | 189          | 150,265 | 128   |
| MNIST; Fashion-MNIST | BVU                              | 157          | 149,945 | 128   |
| Digits               | Plain MLP                        | 330          | 24,760  | 128   |
| Digits               | Confidence MLP                   | 178          | 24,752  | 128   |
| Digits               | Reliability-gated MLP            | 330          | 24,760  | 128   |
| Digits               | BVU                              | 123          | 24,733  | 128   |
| California Housing   | Plain MLP; Reliability-gated MLP | 986          | 9,861   | 128   |
| California Housing   | Confidence MLP                   | 548          | 9,865   | 128   |
| California Housing   | BVU                              | 93           | 9,859   | 128   |
| MNIST; Fashion-MNIST | Same-width Confidence MLP        | 157          | 247,913 | 128   |
| Digits               | Same-width Confidence MLP        | 123          | 17,107  | 128   |

Controlled-reliability rows use the 100% training fraction, missing and Gaussian corruption families, and seeds 0, 1, and 2. They use AdamW, learning rate  $10^{-2}$ , weight decay 0, cross-entropy for classification or MSE for California Housing, a 15,000-step cap, and mean in-distribution accuracy or  $R^2$  for checkpoint selection.

Table 6: Real missing-data experiment settings. For neural models, LR lists the shared predeclared grid; the selected value varied by seed and is recorded in the run files.

| Dataset     | Model                          | Width | Params | Batch | Optimizer | LR                                   |
|-------------|--------------------------------|-------|--------|-------|-----------|--------------------------------------|
| APS         | Plain MLP                      | 579   | 99,589 | 128   | AdamW     | $10^{-3}, 3 \times 10^{-3}, 10^{-2}$ |
| APS         | Confidence MLP                 | 291   | 99,523 | 128   | AdamW     | same grid                            |
| APS         | Same-width Confidence MLP      | 240   | 82,081 | 128   | AdamW     | same grid                            |
| APS         | BVU                            | 240   | 99,601 | 128   | AdamW     | same grid                            |
| APS         | NeuMiss (depths 1, 3, 5)       | 412   | 99,935 | 128   | AdamW     | same grid                            |
| APS         | HistGradientBoosting reference | n/a   | n/a    | n/a   | n/a       | n/a                                  |
| Air Quality | Plain MLP                      | 4,991 | 49,911 | 128   | AdamW     | $10^{-3}, 3 \times 10^{-3}, 10^{-2}$ |
| Air Quality | Confidence MLP                 | 2,773 | 49,915 | 128   | AdamW     | same grid                            |
| Air Quality | Same-width Confidence MLP      | 217   | 3,907  | 128   | AdamW     | same grid                            |
| Air Quality | BVU                            | 217   | 49,911 | 128   | AdamW     | same grid                            |
| Air Quality | NeuMiss (depths 1, 3, 5)       | 4,993 | 50,003 | 128   | AdamW     | same grid                            |
| Air Quality | HistGradientBoosting reference | n/a   | n/a    | n/a   | n/a       | n/a                                  |

APS uses BCE-with-logits loss with the recorded positive-class weight, PR-AUC/average precision for validation checkpoint selection, a 7,000-step cap, and 1,600-step patience (8 checks every 200 steps). Air Quality uses MSE loss,  $R^2$  for selection, a 4,000-step cap, and 1,500-step patience (30 checks every 50 steps). Both use seeds 0, 1, and 2 and fixed dataset splits; no training-data fraction was recorded. The HistGradientBoosting references are non-neural, so neural width, parameter count, batch, optimizer, and LR are not applicable.

## D Supplementary Results

### D.1 Clean supervised-learning benchmark

Table 7 reports the complete clean-data benchmark used in Section 6.1. Results are shown for both the 25% and 100% training-data conditions across all seven datasets.

Table 7: Clean-data test performance of BVU and its parameter-matched MLP on the seven benchmark datasets at 25% and 100% of the training data (mean  $\pm$  sample SD over seeds 0, 1, 2).  $\Delta$  is BVU – matched MLP, computed per seed and then averaged; higher is better for both accuracy and  $R^2$ .

| Dataset            | Train fraction | Metric   | BVU               | Matched MLP       | $\Delta$           |
|--------------------|----------------|----------|-------------------|-------------------|--------------------|
| Breast Cancer      | 25%            | Accuracy | $0.950 \pm 0.025$ | $0.956 \pm 0.015$ | $-0.006 \pm 0.010$ |
|                    | 100%           | Accuracy | $0.968 \pm 0.013$ | $0.968 \pm 0.005$ | $0.000 \pm 0.015$  |
| Wine               | 25%            | Accuracy | $0.963 \pm 0.016$ | $0.963 \pm 0.016$ | $0.000 \pm 0.028$  |
|                    | 100%           | Accuracy | $0.981 \pm 0.016$ | $0.991 \pm 0.016$ | $-0.009 \pm 0.016$ |
| Digits             | 25%            | Accuracy | $0.919 \pm 0.015$ | $0.918 \pm 0.025$ | $+0.002 \pm 0.011$ |
|                    | 100%           | Accuracy | $0.966 \pm 0.007$ | $0.969 \pm 0.008$ | $-0.003 \pm 0.010$ |
| Diabetes           | 25%            | $R^2$    | $0.173 \pm 0.110$ | $0.017 \pm 0.302$ | $+0.156 \pm 0.194$ |
|                    | 100%           | $R^2$    | $0.374 \pm 0.021$ | $0.300 \pm 0.027$ | $+0.074 \pm 0.025$ |
| California Housing | 25%            | $R^2$    | $0.773 \pm 0.013$ | $0.732 \pm 0.022$ | $+0.041 \pm 0.022$ |
|                    | 100%           | $R^2$    | $0.791 \pm 0.005$ | $0.697 \pm 0.079$ | $+0.094 \pm 0.083$ |
| MNIST              | 25%            | Accuracy | $0.961 \pm 0.008$ | $0.943 \pm 0.009$ | $+0.017 \pm 0.015$ |
|                    | 100%           | Accuracy | $0.977 \pm 0.001$ | $0.960 \pm 0.003$ | $+0.017 \pm 0.002$ |
| Fashion-MNIST      | 25%            | Accuracy | $0.856 \pm 0.006$ | $0.833 \pm 0.003$ | $+0.023 \pm 0.005$ |
|                    | 100%           | Accuracy | $0.881 \pm 0.006$ | $0.856 \pm 0.007$ | $+0.025 \pm 0.008$ |

Values are test-set accuracy (classification) or  $R^2$  (regression) at the checkpoint selected on validation data. Both models use the parameter budgets and optimization settings in the reproducibility tables. No significance testing is applied ( $n = 3$  seeds).

## D.2 APS missingness-stratum composition

Because stratified PR-AUC depends on the class composition within each missingness regime, Table 8 reports the number and class balance of APS test examples in each stratum used in Figure 3.

Table 8: Composition of the APS Failure at Scania Trucks test set (official 16,000-example test split) by the fraction of the 170 input features that are missing, using the five strata of Figure 3. Bins are upper-inclusive and contain identical examples for every seed.

| Missing fraction | Test examples | Positive | Negative | Positive fraction |
|------------------|---------------|----------|----------|-------------------|
| 0                | 165           | 30       | 135      | 0.182             |
| (0, 0.10]        | 11,981        | 132      | 11,849   | 0.011             |
| (0.10, 0.25]     | 2,970         | 79       | 2,891    | 0.027             |
| (0.25, 0.50]     | 770           | 129      | 641      | 0.168             |
| > 0.50           | 114           | 5        | 109      | 0.044             |
| All              | 16,000        | 375      | 15,625   | 0.023             |